

A ZFC Answer To A Steinhaus Property Renorming Question

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Status

This is a **literature-already-answered** packet. It records a later paper that explicitly answers one question left open in Kania–Kochanek [1]. No original proof is claimed here.

Source Question

The source paper is Kania–Kochanek, arXiv:1302.6443 [1]. In the introduction, after Theorem B, the authors explain that their example of a Banach space with property (S) but with no equivalent strictly convex norm assumes that the continuum is a real-valued measurable cardinal. They then write that it remains open whether such constructions can be obtained in ZFC.

Supporting Answer

The supporting paper is Johnson–Kania, *Uniform Property (S)* , arXiv:2602.09106 [2]. In its introduction, immediately before and after Corollary 1.5, it states that Theorem A positively answers the question left open in [1] about the existence, in ZFC, of spaces with property (S) that do not admit equivalent strictly convex norms.

The mechanism is that for every Banach space X , the Bochner space

$$L_1([0, 1], X)$$

has uniform property (S) , hence property (S) , and contains X isometrically as the constant functions. Choosing X to be a Banach space with no equivalent strictly convex renorming, such as $\ell_\infty(\Gamma)$ for uncountable Γ or ℓ_∞/c_0 , gives in ZFC a Banach space with property (S) and no equivalent strictly convex norm.

Scope Limitation

This packet is only about the ZFC existence question for property (S) spaces that are not strictly convexifiable. It does *not* answer the separate question, also raised in [1], whether every Banach space admits an equivalent norm satisfying property (F) , meaning that the unit sphere has no non-empty relatively open subset contained in a hyperplane.

References

- [1] T. Kania and T. Kochanek, *Steinhaus' lattice-point problem for Banach spaces*, J. Math. Anal. Appl. 446 (2017), 1219–1229; arXiv:1302.6443.
- [2] W. B. Johnson and T. Kania, *Uniform Property (S)*, arXiv:2602.09106.